

Predictive Maintenance

Stress-testing the effect of extreme environmental conditions on the ___ of UGVs after 90 days

Rough:

Inputs

Extreme heat (Middle East) degrades engine performance

Extreme cold (Eastern Europe) kills batteries and hydraulics

Dust ingestion - filters clog, engines overheat

Humidity - moisture gets into connectors

Salinity - accelerates corrosion on everything metal, destroys electrical contacts. Iran has salt flats.

Ukraine has the Black Sea coast.

Solar radiation / UV - rubber seals and gaskets shrink, optical coatings on cameras, rangefinders, and periscopes degrade

Output

Part failure - which part fails first (info about that part like where it comes from)

Sensor reliability - radar, thermal optics, acoustic sensors all degrade differently under humidity, dust, and temperature swing

Battery degradation

Hydraulics

Filters

Corrosion

Final

Inputs

Extreme temperature

Dust ingestion

Humidity
Salinity
UV / solar radiation

Outputs

Detailed

Environmental Factors (inputs)

- I. Extreme heat
 - A. Degrades engine performance
- II. Extreme cold
 - A. Kills batteries and hydraulics
- III. Dust ingestion
 - A. Clogs filters → engines overheat
- IV. Humidity
 - A. Moisture gets into connectors → corrosion on metal contacts → heat buildup
- V. Salinity
- VI. UV / solar radiation

Critical Components (outputs)

- I. Chassis
 - A. Frame
 - B. Plating
 - C. Suspension
 - D. Underbelly
 - E. Track / wheels
 - F. Hatches & doors
- II. Sensors
 - A. Thermal
 - B. Radar
 - C. Acoustic
 - D. GPS
 - E. Camera

- III. Battery
- IV. Engine
- V. Hydraulics